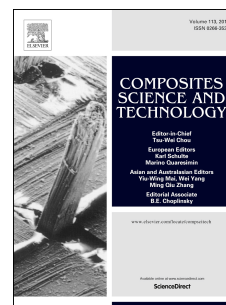


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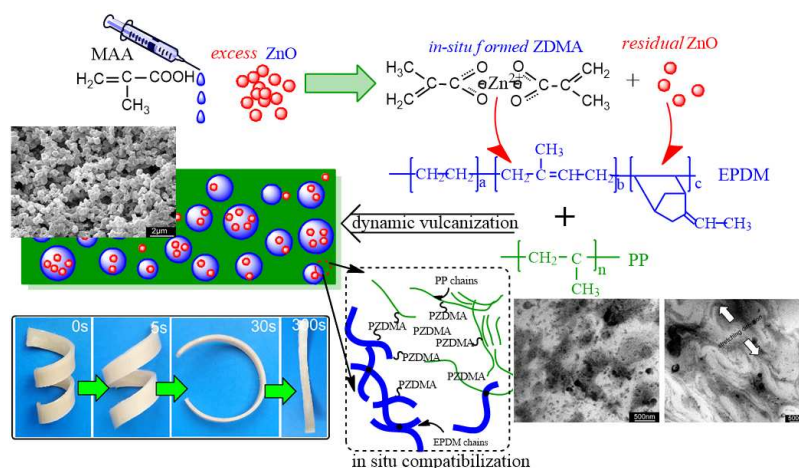
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Graphical Abstract



Design of shape-memory materials based on sea-island structured EPDM/PP TPVs via in-situ compatibilization of methacrylic acid and excess zinc oxide nanoparticles

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Abstract

It is a challenge to achieve considerable shape-memory (SM) effect for typical sea-island structured polymer blends. In this paper, we successfully realized the SM behavior for classical ethylene-propylene-diene rubber/polypropylene (EPDM/PP) thermoplastic vulcanizates (TPVs) through a strong rubber/plastic interface compatibilized by in-situ formed zinc dimethacrylate (ZDMA) which came from methacrylic acid (MAA) and excess zinc oxide (ZnO) nanoparticles. With MAA/ZnO, the average size of EPDM particles reduced to 400~500 nm, which significantly enlarged EPDM/PP interfacial contact surfaces. The enhanced interface improved the efficiency of stress delivery between PP and EPDM phases, which was critical to fulfill the SM behavior. At the same time, residual ZnO nanoparticles stressed a reinforcement on EPDM phase and PP phase, which further improved the shape recovery (SR) and shape fixing (SF) of TPVs. Enhanced tensile strength (~9.1 MPa) and Young's modulus (~380 MPa), improved SF (~99%) and SR (~98%) of TPVs were achieved by tailoring the content of residual ZnO nanoparticles (MAA/ZnO=2:1.3) and the shape deformation temperature (120 °C).

Keyword: A. Polymer-matrix composites (PMCs); B. Interface; C. Deformation; shape memory

1. Introduction

Shape-memory polymers (SMPs) are a kind of promising smart materials capable to memorize temporary shapes and recover their initial shapes when suffered external stimulations [1]. In light of the intelligent memorizing capability, SMPs have great potential applications in smart devices, packaging, sensors and actuators [2]. Among various stimulations, heat is the most convenient way to trigger the shape-memory (SM) behavior [3]. To earn the charming SM behavior, heat-induced SMPs are usually designed to contain at least two essential functional components [4,5]: one is called as switching phase with the responsibility to maintain the temporary shape, the other one is the reversible phase which is easily deformed to provide driving force for recovery. The glass (T_g) or melting transition (T_m) can be served as switching temperature (T_s) to fix or trigger the temporary shape of SMPs [6-8]. Along with T_g and T_m , anisotropic/isotropic transition (T_i) in liquid crystalline networks [9-10], reversible molecule cross-linking [11] and supramolecular association/disassociation [12] can also be serve as switches to trigger SM behaviors.

Compared with complicated chemical synthesis [13], polymer blending has been receiving growing focuses on fabrication of SMPs due to its convenience, availability and efficiency in preparation [14-17]. In polymer blending systems, the hard domains can be frozen/unfrozen upon crystallization/melting to fix/trigger the temporary shape while the soft domains provide elastic restoring force to recover the shape [18]. There have been tremendous reports on polymer blends showing SM effect based on PLA/PU [19], PLA/NR [20], PLA/ENR [21], PLA/EGVA [22], PLA/PPC [23], PLA/PA [24], PLA/PMMA [25], PLA/PCL [26], PLA/PVAc [27], PCL/EVA [28], etc. However, most of them contain at least one main component owning apparent SM characteristics, e.g. the PLA and PCL [29-32]. The other blending components assist the main component to fulfil better SM behavior.

So, how about other polymer blending systems? In fact, several reports have substantiated that SM behavior can also be realized in a co-continuous phase structure in which one component can be any amorphous or crystalline polymer serving as switch phase and another component can be elastomer with reversible mobility [33]. Actually, a co-continuous structure with high elastomeric phase continuity would provide strong elastic restoring force and improve the efficiency of stress delivery [34,35]. For example, Zheng et.al [36] prepared TPU/SLB multilayer materials consisting of co-continuous PCL and PBS in SLB through layer-multiplying coextrusion. It was reported that shape fixity and recovery ratios were beyond 95 and 85% in dual- and triple-SM processes, respectively, which were difficult to be achieved in other morphology structures [36]. Unfortunately, it is not easy to control the formation of co-continuous phase structure in an immiscible blending system through conventional processing method, especially in plastic/rubber thermoplastic vulcanizate (TPV) systems [37]. The phase separation turns out a sea-island structure in which the crosslinked rubber particles are dispersed in plastic continuous phase [38,39].

Therefore, it is more practicable and challengeable to achieve considerable SM behavior for a typical sea-island structured polymer blend. Compared with continuous elastomeric phase, dispersed elastomer domains in sea-island morphology provide inferior elastic restoring force. Moreover, the phase interface is critical to SM behavior since that the stress delivery from dispersed elastomer phase to continuous plastic phase directly determined the shape recovery [40]. The poor interface is easily destroyed during shape deformation, which interrupts the stress

delivery efficiency and eliminates potential shape recovery. Obviously, crosslinking of elastomer domains to earn stronger elastic restoring force and enhancing interface to improve stress delivery are two key factors for SM effect of sea-island structured polymer blends.

We focus on a famous ethylene-propylene-diene rubber/polypropylene (EPDM/PP) TPV system. Considering the substantial market shares and cost advantage of commercial EPDM/PP TPVs, the successful SM functionalization might open up new potential applications for them. In our previous work [41], we employed zinc dimethacrylate (ZDMA) as compatibilizer for EPDM/PP TPV. The effective EPDM/PP weight ratio was found to be 70/30 where high rubber content could provide sufficient elastic restoring force to drive the shape recovery. However, the drawback of such a high rubber-rich EPDM/PP ratio was that the thin PP continuous phase was easily crazed during fixing temporary shape. If the EPDM/PP ratio reached 80/20 and 90/10, the poor shape fixity completely eliminated the SM effect of TPVs. To overcome this drawback, in this paper, in-situ reaction of methacrylic acid (MAA) and excess zinc oxide (ZnO) was employed to replace ZDMA and the EPDM/PP ratio was fixed at 60/40 to strengthen the PP phase. The in-situ compatibilization of MAA/ZnO also produced a strong EPDM/PP interface to realize SM behavior for TPVs. Many reports have verified that the reinforcement of MAA/ZnO on rubbers is superior to that of ZDMA [42-45], thus it provides more returns of shape driving force to make up for the reduced EPDM content. More importantly, on the one hand, the residual excess ZnO nanoparticles added additional reinforcement on EPDM particles to further improve the shape recovery, on the other hand, ZnO nanoparticles served as heterogeneous nucleation agent for PP phase to increase crystallinity which contributed to shape fixing. As expected, the final TPV exhibited improved SM behavior and mechanical properties.

2. Experimental section

2.1 Materials.

PP was purchased from CNOOC & Shell Petrochemicals Company Limited, its T_m was about 167 °C and MFI was 12 g/10min (210 °C, 2.16 kg); EPDM, with 56% ethylene and 7.5% 5-ethylidene-2-norbornene, was purchased from Jilin Chemical Co. (China), its density was 0.87 g/cm³ and ML (1+4, 100 °C) was 40. Dicumyl peroxide (DCP), purchased from Guangzhou Xingang Chemic factory (China), was purified by anhydrous alcohol recrystallization before use. ZnO (analytical pure) was purchased from XiLong Chemical Company (Guangzhou, China), the average size was about 50 nm. MAA (analytical pure), was purchased from Sinopharm Chemical Reagent Co. Ltd (China).

2.2 Preparation of EPDM/MAA/ZnO compounds.

In-situ reaction of MAA and ZnO was mainly completed during mastication of EPDM in a two-roller mill. ZnO nanoparticles were first blended into EPDM for 5 min until achieving a homogeneous dispersion. Then, MAA was carefully instilled into EPDM/ZnO compound using an injector. The temperature of rollers was kept at 45~50 °C to help the reaction of ZnO and MAA. This process was about 8 min. To further promote the neutralization of MAA and ZnO, the EPDM/MAA/ZnO compounds were then transferred into the mixing chamber of a Haake Rheocord 90 and mixed for another 5 min at the rotor speed of 32 rpm and the temperature of 60 °C. More details about the in-situ formation of ZDMA via neutralization of MAA and ZnO during mastication can be found in references [43-46]. According to the molecular formula of

ZDMA that one Zn^{2+} combined two $-\text{OOC}-\text{C}(\text{CH}_3)=\text{CH}_2$, the theoretical idea molar ratio of MAA:ZnO was 2:1. So, we controlled the MAA load to set the weight ratio of EPDM/ZDMA=100/20 and selected a series of MAA/ZnO molar ratio of 2:1, 2:1.1, 2:1.2 and 2:1.3 to make ZnO nanoparticles were excess. The high mixing temperature, the long mixing time and excess ZnO should make ZnO and MAA react to their utmost to form ZDMA [44,46].

2.3 Preparation of EPDM/PP TPVs.

In this paper, EPDM/PP weight ratio was fixed at 60/40. Dynamic vulcanization of TPV was done in Haake Rheocord 90 at a rotor speed of 50 rpm and initial temperature at 170°C. After PP was shear-melted completely, EPDM/MAA/ZnO compound was added. About 3 minutes later, DCP (1.5 wt% of EPDM component) was added to initiate the crosslinking of EPDM and the polymerization of MAA/ZnO. The mixing was continued until to the final stable torque. The whole time was controlled at 10 min. The TPVs were then removed from the chamber and cooled down to the room temperature. For simplicity, TPVs were named according to the MAA/ZnO molar ratio, e.g. E6P4-2:1.2 represented a TPV (EPDM/PP=60/40) with a MAA/ZnO molar ratio=2:1.2.

2.4 Characterizations.

Dynamic mechanical analysis was determined on a DMA242C (NETZSCH, Germany) under tensile mode. Test samples were rectangular shape with 8 mm × 4 mm × 0.5 mm. Frequency was set at 1 Hz and heating rate was 3 °C /min. The scanning temperature was ranged from -100 to +100 °C.

The morphologies of TPVs were observed in a Nova NanoSEM 430 (FEI Company, USA). To observe the crosslinked EPDM phase, TPVs were etched by hot xylene wash to remove the PP phase at the surface and then dried completely. Before observation, the surface of sample was treated by spray-gold.

TEM observations were carried out on a JEM-2100F operating at 200 kV. The specimens were ultramicrotomed into thin sections using a Leica EM UC6 equipment under liquid nitrogen atmosphere.

Morphological observations on crystallites of the PP phase were conducted on an Axioskop 40 Polarized optical microscopy (POM) (ZEISS, Germany) equipped with a hot stage under crossed polarizers. The samples were first melted at 200 °C on the hot stage and held for 5 min to achieve thermal equilibrium. Then, it was rapidly cooled to 150 °C and isothermally crystallized at this temperature. The POM micrographs were recorded.

Crystallinity of PP phase was determined using a DSC 240 F1 (NETZSCH, Germany). Testing program: (1) heating from room temperature to 200 °C at a rate of 20 °C /min and hold at 200 °C for 5min to eliminate heat history; (2) cooling down to 30°C at a rate of 10°C/min; (3) reheating to 200 °C at a heating rate of 20 °C/min. Then, the crystallinity (X_c , %) was determined according to equation:

$$X_c = \frac{\Delta H_c}{\Delta H_c^* \phi} \times 100\% \quad (1)$$

where the heat of crystallization (ΔH_c) was determined from the area under the exotherm, the heat of fusion of 100% PP was 209 J/g and ϕ is the mass fraction of PP in the TPV.

Stress-strain behaviors at room temperature were carried out on a UT-2080 (U-CAN, Taiwan) under a tensile mode at a crosshead speed of 500 mm/min. Tensile tests at higher temperatures (100~140 °C) were carried out in an Instron 5500R machine. TPVs were first achieved equilibrating at target temperature for 10 min and then stretched at 50 mm/min. The thicknesses of the dumbbell shaped sample sheets were 1mm. Tensile tests at elevated temperature were repeated 5 times to ensure repeatability.

SM properties were tested according to followed five stages: (a) marking the length of 2 cm at the center of a dumbbell specimen by two distance lines (L_0); (b) allowing the specimen at deformation temperature for ~10min to achieve thermal equilibrium; (c) applying a load to elongate marked 2 cm to 3 cm and marking distance as L_1 ; (d) cooling down the elongated specimen to room temperature under loading, unloading and 24 h later, recording the marked distance as L_2 ; (e) placing the specimen at recovery temperature, 300s later, recording the marked distance as L_3 . Shape fixity ratio (SF) and shape recovery ratio (SR) were calculated according to the followed equations. Every sample was measured for 5 individual specimens to achieve the average value.

$$SF = \frac{L_2 - L_0}{L_1 - L_0} \times 100\% \quad SR = \frac{L_2 - L_3}{L_2 - L_0} \times 100\% \quad (2)$$

Stress-controlled cyclic thermomechanical experiments were carried out by using DMA Q800 instrument to evaluate the SM behaviors of EPDM/PP TPVs. Test samples were tailored to be rectangular shape with the size of 8.0 × 6.0 × 1.0 mm. Before launching the SM cycles, samples were annealed at 120 °C for 10 min to eliminate the stress history. Then, the SM cycles were done according to four stages as followed [22]: (a) stretching to 100% strain by a linearly increased stress with a rate of 0.5 N/min (preloaded value was 0.005 N). (b) cooling to 30 °C under the loading force. (c) unloading the applied force to preloaded value at a rate of 0.5 N/min and holding at 30 °C for 10 min. (d) heating to 165 °C at a rate of 3 °C/min and holding at 165 °C for 10 min. The above cycle was repeated for 2 times.

3. Results and discussion

3.1 Strategy for realizing SM effect of EPDM/PP TPVs

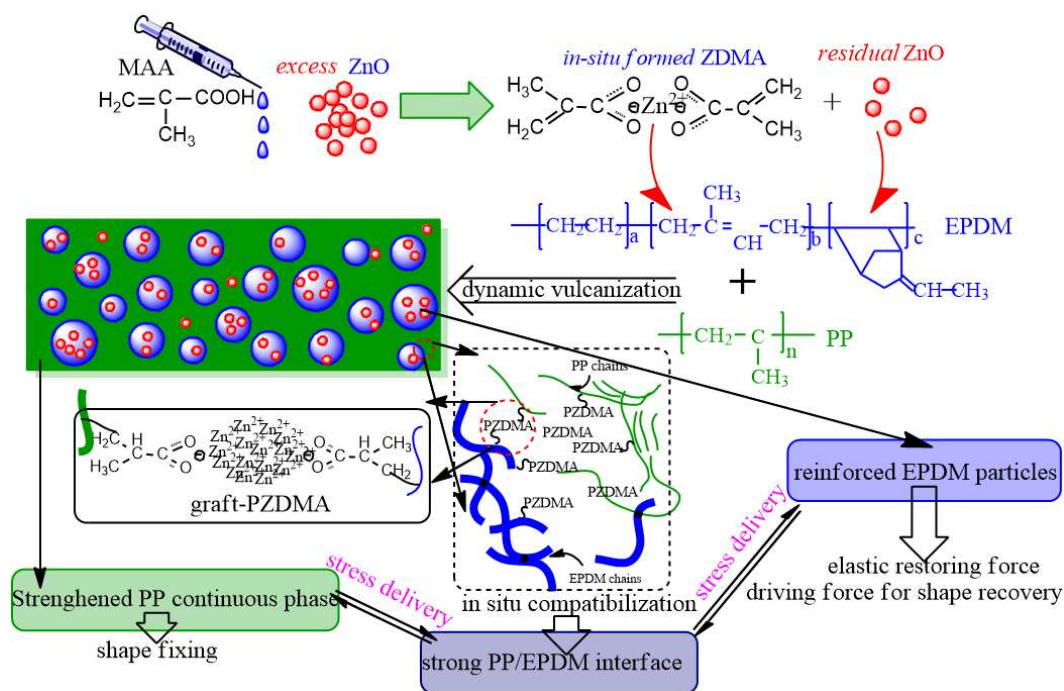


Figure 1 Schematic illustration of the strategy for realizing SM effect of EPDM/PP TPVs via in situ compatibilization of MAA and excess ZnO nanoparticles

The design idea of EPDM/PP TPVs is schematically illustrated in Figure 1. The key feature of our approach to realize SM effect for EPDM/PP TPVs is to generate a strong rubber/plastic interface to improve the efficiency of stress delivery. Our strategy is based on the in-situ compatibilization of in-situ formed ZDMA from MAA and excess ZnO nanoparticles. In light of the high reaction activity, ZDMA is readily graft-polymerized with EPDM and PP during DCP-induced dynamic vulcanization, which results in a powerful EPDM/PP interface [47]. The graft polymerization of ZDMA [48] and the excellent reinforcement of in-situ formed ZDMA on rubbers have been reported in many documents [42-45], and, the detailed FTIR characterization of polymerized ZDMA (PZDMA) compatibilized EPDM/PP interface can be found in our previous reports [41,47]. Because the ZnO is excess, the residual ZnO nanoparticles yield an additional reinforcement on EPDM phase [45]. After dynamic vulcanization, the crosslinked EPDM particles provide strong elastic restoring force to drive the shape recovery, which makes up for the reduced EPDM content (EPDM/PP=60/40 in this paper). Furthermore, the increased PP proportion turns out perfect shape fixing, which overcomes the drawback that the thin PP continuous phase was easily crazed during fixing temporary shape in our previous work (EPDM/PP=70/30).

3.2 Dynamic vulcanization, size reduction of crosslinked EPDM particles and in-situ compatibilization of MAA/ZnO

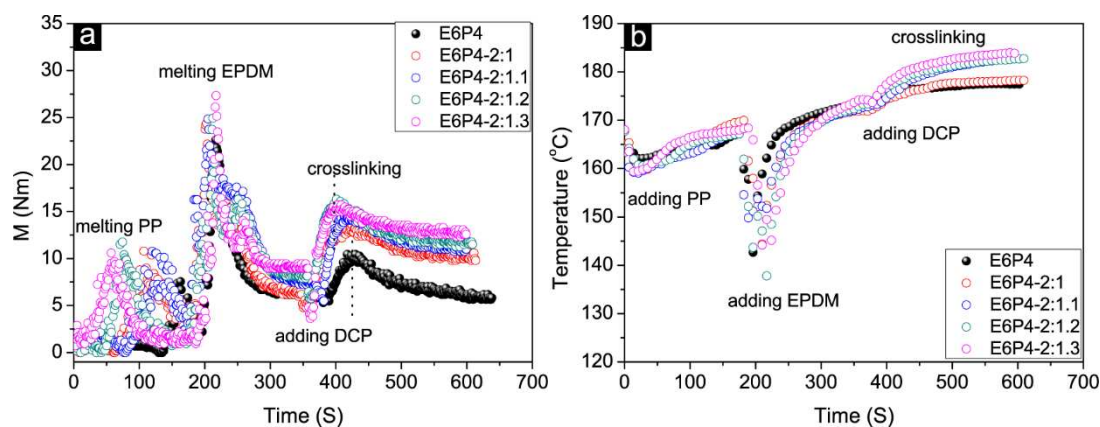


Figure 2 (a) Torque changes and (b) temperature monitoring during dynamic vulcanization

Figure 2a shows the torque changes during dynamic vulcanization of TPVs. The first torque peak corresponds to the melting of PP pellets and the second torque peak is attributed to the addition and fragmentation of EPDM/MAA/ZnO compound. The torques were stable at 5~6 min, which indicated that the blending had achieved full homogenization. Afterward, DCP was added to induce the dynamic vulcanization and the polymerization of MAA/ZnO. As seen, crosslinking led to an abrupt increase in torque at about 7 min due to the drastic increase in the viscosity and elasticity of EPDM phase [38]. It is worth noting that the torque slope of crosslinking peak increases with the increasing ZnO, which indicates that the residual ZnO nanoparticles promoted the crosslinking reaction of EPDM phase. For example, compared with the E6P4-2:1, the time reaching crosslinking peak of E6P4-2:1.3 was shortened ~30 s. According to our previous report [47], the decomposition of DCP induced EPDM, PP and PZDMA macroradicals via the hydrogen abstraction, which conducted the in-situ compatibilization at phase interfaces. Besides, the promoted crosslinking of EPDM domains by MAA/ZnO facilitated breaking down to smaller sizes during dynamic vulcanization [41], which turned out a higher final torque. As a result, the increased friction between EPDM particles and PP phase raised the temperature in the mixing chamber (Figure 2b). The effect of ZDMA on the promoting the crosslinking of rubbers can be found in reference [48].

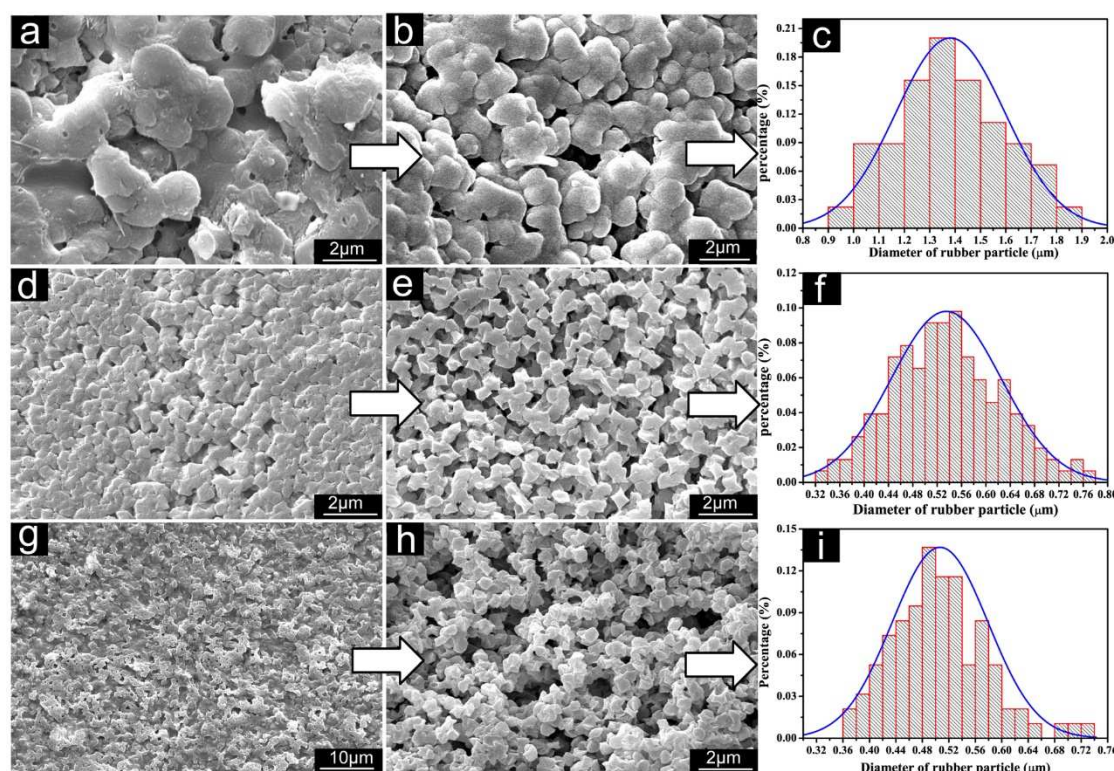


Figure 3 SEM images of TPVs etched by xylene at 100 °C for 3 min and 20 min and their corresponding size distributions: (a, b, c) E6P4; (d, e, f) E6P4-2:1; (g, h, i) E6P4-2:1.3

The size reduction of EPDM particles was confirmed from SEM observation. In order to get a clear understanding of the dispersion of EPDM particles, we controlled the etching time (TPVs etched by xylene at 100 °C for 3 min) to remove partial PP phase on the surface layer of TPVs. As seen, the semi-exposed particles dispersed uniformly in the continuous PP phase, which was benefit to the stress delivery during deformation of TPVs (Figure 3a, 3d and 3g). After further removing the free PP (TPVs etched by xylene at 100 °C for 20 min), the EPDM particles in E6P4 showed a decreased size (Figure 3b). Compared with the E6P4 (average size was about 1.4~1.5 μm, Figure 3c), the average size of EPDM particles in E6P4-2:1 was reduced to about 500nm (Figure 3e). ZDMA promoted the crosslinking rate of EPDM and rapidly increased its viscosity and elasticity at the initial stage of dynamic vulcanization [47,49]. At the same time, the in-situ compatibilization at EPDM/PP interface increased the interactions between PP melt and EPDM phase. All these facilitated the EPDM domains being sheared into smaller sizes. As for the E6P4-2:1.3, a further reduced average particle size was achieved (Figure 3h). Besides, its corresponding size distribution became narrow (comparing Figure 3f and 3i), which enlarged the EPDM/PP interfacial contact surfaces. This might be attributed to that the residual ZnO nanoparticles further promoted the crosslinking of EPDM and hence improved the shearing efficiency.

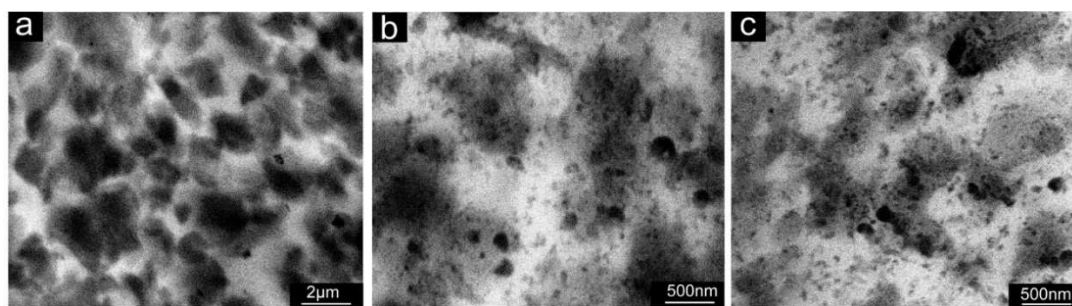


Figure 4 TEM images of (a) E6P4; (b) E6P4-2:1; (c) E6P4-2:1.3

The phase structure of TPVs was determined by TEM. As shown in Figure 4a, the light phase represents the PP and the dark particles correspond to the EPDM. The boundary of EPDM particles is relative clear in E6P4. However, it becomes obscured in E6P4-2:1 (Figure 4b) and E6P4-2:1.3 (Figure 4c). Besides, a large number of nanoparticles with size about 20~50nm appear in E6P4-2:1 and E6P4-2:1.3, which are associated with the ZnO and PZDMA [50,51]. Unfortunately, it is difficult to distinguish ZnO and PZDMA since their sizes are similar considerably. Nevertheless, the blurry boundary strongly confirmed that the compatibility between PP phase and crosslinked EPDM particles was improved [41]. As a result, the strengthened interface combined with the increased EPDM/PP contact surfaces provided essential conditions to realize the SM effect for TPVs. The sizes of EPDM particles observed in TEM agreed well with the SEM results.

3.3 Dynamic mechanical properties

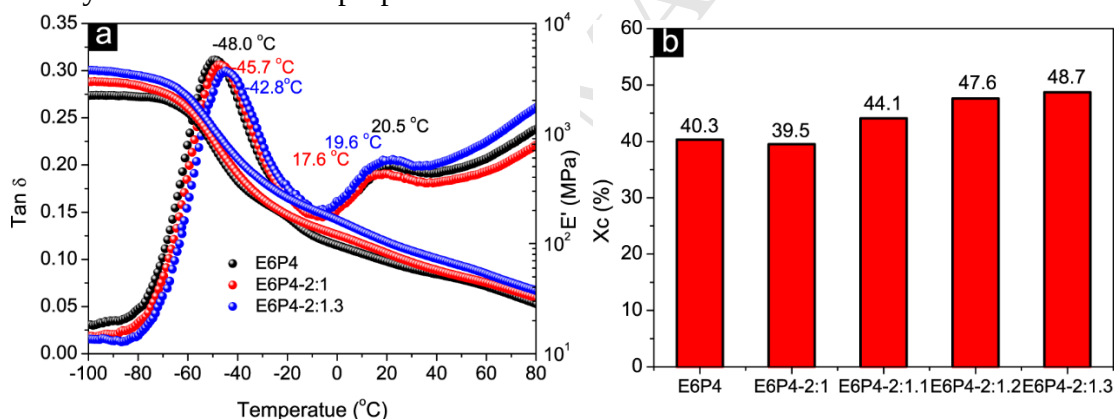


Figure 5 (a) Loss factor ($\tan \delta$) and storage modulus (E'); (b) crystallinity (X_c %) of TPVs

The shift of $\tan \delta$ peak temperature is used to confirm the raised crosslink density of EPDM phase. As shown in Figure 5a, with MAA/ZnO (2:1), the $\tan \delta$ peak of EPDM phase was shifted from -48 to -45.7 °C while it was decreased from 20.5 to 17.6 °C for PP phase. The $\tan \delta$ peak of EPDM shifting to higher temperature was due to the decrease of chain mobility imparted by the increased crosslink density from polymerization of MAA/ZnO [41], while the $\tan \delta$ peak of PP shifting to lower temperature further confirmed the successful compatibilization. As for the E6P4-2:1.3, the residual ZnO nanoparticles further restrict the chain mobility [52,53] of EPDM, which turned out a higher temperature of $\tan \delta$ peak (-42.8 °C). It is well known that a higher crosslink density is benefit to improve the mechanical properties of rubbers. Therefore, the EPDM particles in E6P4-2:1.3 processed stronger shape driving force. Interestingly, the $\tan \delta$ peak of PP phase also showed a shift toward high temperature (19.6 °C). This could be explained as the heterogeneous nucleation effect of migrated ZnO nanoparticles which increased the crystallinity of PP [54] (Figure 5b). The heterogeneous nucleation effect of ZnO nanoparticles was confirmed

by polarized optical microscope observation (Supporting Information Figure S1). Compared with E6P4 (Figure S1a), E6P4-2:1 (Figure S1b) exhibited reduced size of spherulites. The size reduction of EPDM particle resulted in higher particle density, which had two conflict effects on the crystallization of PP phase [41,55]: hindering crystal growth by increased particle numbers and providing more platforms for the heterogeneous nucleation by increased specific surface of EPDM particles. As a result, the spherulite impingement on each other reduced the spherulite size. The increased ZnO nanoparticles in PP phase served as nucleating agents, which promoted the crystallinity but further reduced the spherulite size, as shown in Figure S1c~e. The increased crystallinity in PP phase was beneficial to the shape fixing of TPVs [56].

3.4 Mechanical properties

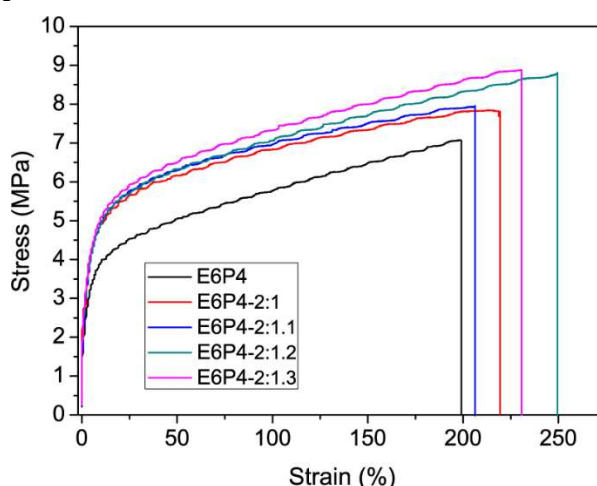


Figure 6 Typical tensile curves of TPVs obtained at room temperature

Therefore, the reinforced EPDM particles, compatibilized EPDM/PP interface and increased crystallinity of PP phase should be considered to explain the increased E' (Figure 5a) and the improved tensile behaviors (Figure 6). The mechanical properties are provided in Table 1. As seen, the in-situ compatibilization of MAA/ZnO (2:1) significantly improved the tensile behavior of TPV while the residual ZnO nanoparticles made it further improved. For example, the Young's modulus was increased from 250 MPa (E6P4) up to 340 MPa (E6P4-2:1) after in-situ compatibilization, and it further increased to 380 MPa (E6P4-2:1.3) when the MAA/ZnO ratio increased to 2:1.3. This reinforcement of residual ZnO nanoparticles on TPVs is favorable to achieve a better shape fixing, and, the enhanced E' facilitated PP phase forcing shape deformation of the EPDM particles and hence storing sufficient elastic restoring force for shape recovery.

Table 1. Mechanical properties of TPVs (5 specimens for each sample).

Samples	MAA/ZnO ratios				
	0	2:1	2:1.1	2:1.2	2:1.3
Tensile strength (MPa)	7.0±0.5	7.8±0.2	8.0±0.2	8.8±0.4	9.1±0.6
Elongation (%)	200±22	220±30	210±25	250±31	230±45
Young' modulus (MPa)	250±20	340±25	352±22	365±30	380±30

3.5 SM behaviors of EPDM/PP TPVs and SM mechanism analysis

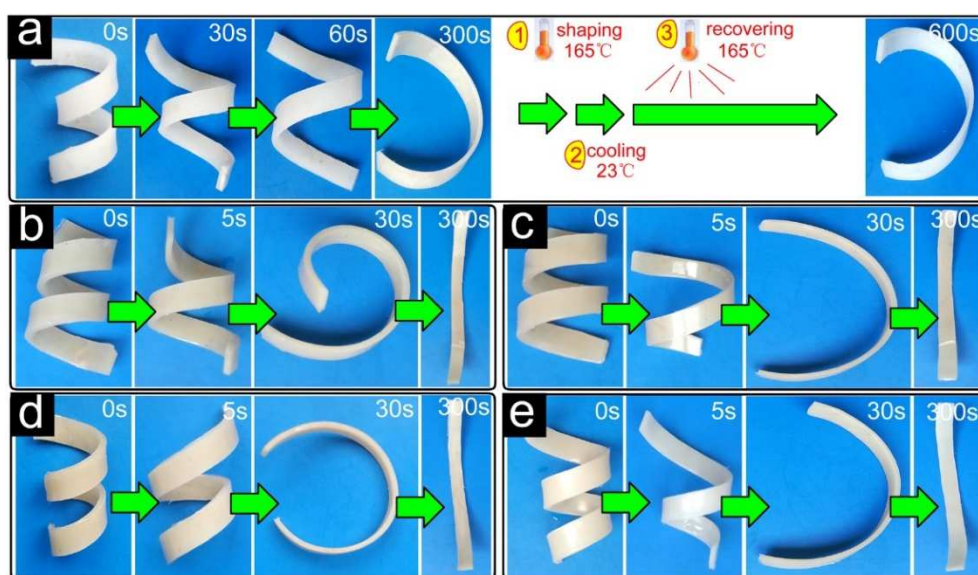


Figure 7 Snapshots of the shape recovery of spiral shaped TPVs: (a) E6P4; (b) E6P4-2:1; (c) E6P4-2:1.1; (d) E6P4-2:1.2 and (e) E6P4-2:1.3

Snapshots of the shape recovery of spiral shape are shown in Figure 7 to give a direct cognition of the SM effect of TPVs. To facilitate the shaping, deformation temperature (T_d) was selected as 165 °C where the TPVs were softened by melting of majority of PP crystals. Similar to other reported SMPs, increasing temperature could reduce the shape recovery time. However, above the melting point of PP, the TPVs became sticky quickly and lost its shape due to that the PP phase was melted completely. Therefore, the trigger temperature far beyond T_m of PP phase is not suitable for this system. The trigger temperature of TPVs in this study was also fixed at 165°C. As seen from Figure 7a, the E6P4 still retained a high residual strain even after being placed at 165°C for 10 minutes. However, compatibilized TPVs showed an apparent SM effect that it only took 5 minutes to complete more than 80% recovery for the E6P4-2:1 (Figure 7b), E6P4-2:1.1 (Figure 7c), E6P4-2:1.2 (Figure 7d) and E6P4-2:1.3 (Figure 7e). This comparison strongly suggested that the in-situ compatibilization of MAA/ZnO was key to the SM functionalization of EPDM/PP TPVs. However, it was a pity that the compatibilized TPVs were unable to fully recover to their original shapes. This was attributed to the high T_d where the PP continuous phase lost most of its strength and could not apply effective force on deforming the EPDM particles. As a result, the slightly deformed EPDM particles were unable to stress sufficient driving force on shape recovery.

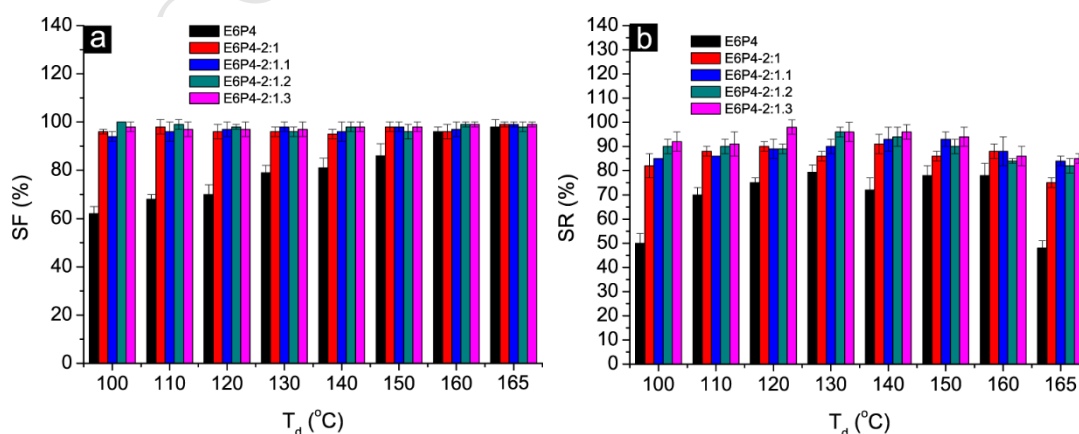


Figure 8 SM effect of TPVs obtained at various T_d : (a) SF and (b) SR

Obviously, besides the important function of powerful EPDM/PP interface in deforming and keeping the deformed EPDM particles in the temporary shape, the strength of PP phase at T_d were quite important in stressing the deformation of EPDM particles. We determined the SF and SR of TPVs obtained at various T_d and show them in Figure 8a and 8b, respectively. All the compatibilized TPVs showed both remarkably improved SF and SR compared with the E6P4. The SF of compatibilized TPVs was maintained above 90% at the whole range of T_d , which suggested that the PP continuous phase was not ruptured during deformation. However, the SF of E6P4 obtained at T_d below 160°C were not ideal. The typical photographs of TPVs shaped at various T_d are also shown in Figure S2 (Supporting Information). As seen, the E6P4-2:1 and E6P4-2:1.3 had excellent shape fixing, while the E6P4 showed obvious shape retractions due to the crazed PP phase initiated by the poor EPDM/PP interface. To address this, SEM observation was done on the longitudinal stretching surface of TPVs at their temporary shapes ($T_d = 120^\circ\text{C}$). As shown in Figure 9a, lots of folds appears on the longitudinal stretching surface of E6P4, which was originated from the separated phases slipping at EPDM/PP interfaces. The detailed structure in the inset of Figure 9a confirmed the crazed PP phase, which should answer for the low SF of E4P6 ($T_d = 120^\circ\text{C}$). However, no granular substances are found in Figure 9a, this implied that the EPDM particles were elongated in E6P4, which turned out a 70% SR (Figure 8a). As for the E6P4-2:1 (Figure 9b) and E6P4-2:1.3 (Figure 9c), no cracks and no EPDM particles were observed on the longitudinal stretching surface. The strengthened PP phase maintained its integrity, which provided sufficient rigidity to against the resilience of deformed EPDM particles. After treatment with hot xylene steam, numerous particles appeared on the longitudinal stretching surface of E6P4-2:1.3 (Figure 9d), which was associated with the EPDM particles. Without the restriction of PP continuous phase, the elongated EPDM particles recovered rapidly to their granular shapes.

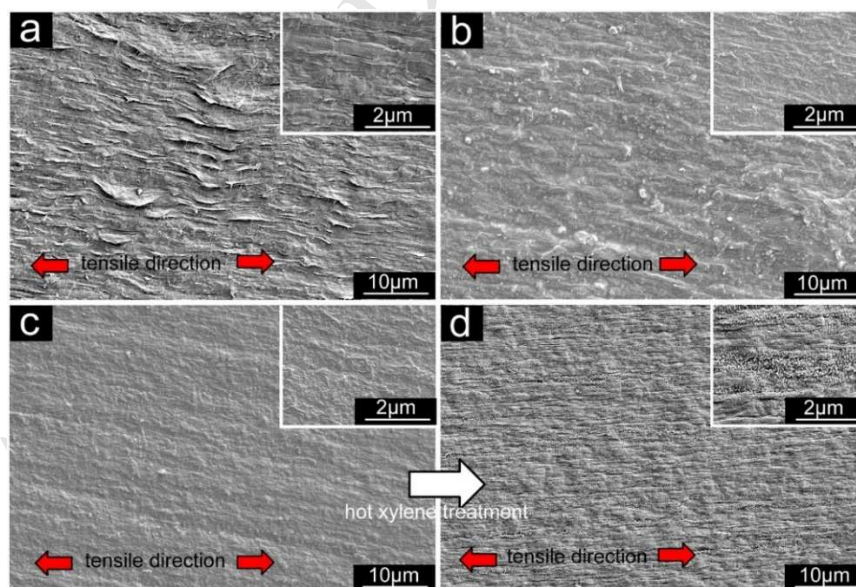


Figure 9 SEM images of longitudinal stretching surface of (a) E6P4; (b) E6P4-2:1; (c) E6P4-2:1.3 and (d) E6P4-2:1.3 treated by hot xylene steam

Thus, the SR driving force of TPVs was mainly originated from the resilience of deformed EPDM particles which were firmly attached with PP continuous phase through the enhanced interface. As discussed previously, the strengthened PP phase could stress more power on

deforming EPDM particles via the compatibilized EPDM/PP interface, at the same time, the reinforced EPDM particles under elongated status feedbacked stronger resilience driving force to shape recovery. To address this, the TPVs were conducted stretching at 120 °C and their stress-strain curves are shown in Figure S3 (Supporting Information). The improved tensile behavior confirmed that the reinforcement of residual ZnO nanoparticles still worked well at high temperatures. Moreover, the largely increased EPDM/PP contact areas also contributed to the stress delivery. As a result, the SR of compatibilized TPVs achieved above 80% in range from 100~160 °C, as shown in Figure 8b. As expected, the residual ZnO nanoparticles further improved the SR. For example, the E6P4-2:1.3 achieved a SR of 98%, 96%, 96% and 94 when deformed at 120, 130, 140 and 150 °C, respectively, showings excellent SM effect. The typical photographs of shape recovery of E6P4, E6P4-2:1 and E6P4-2:1.3 obtained at various T_d are provided in Figure S4 (Supporting Information).

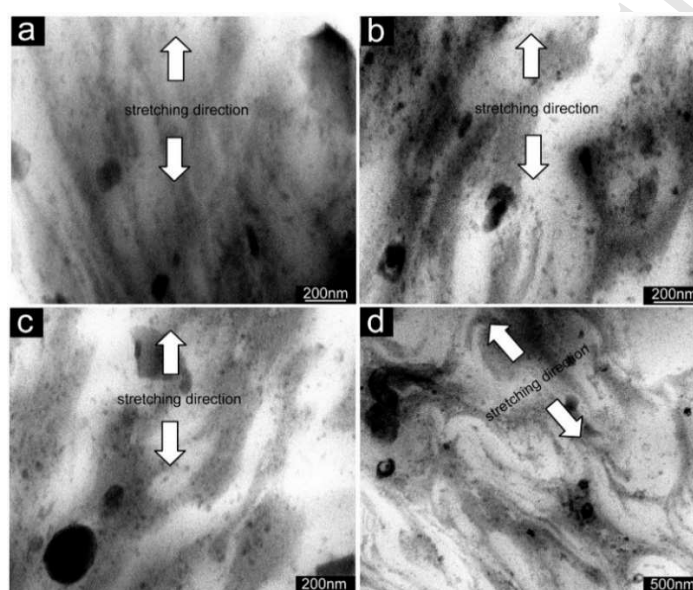


Figure 10 TEM observations at section along the longitudinal deforming direction: (a) E6P4-2:1; (b) E6P4-2:1.1; (C) E6P4-2:1.2; (D) E6P4-2:1.3

In an effort to further understand the essential SM mechanism, the middle part of TPVs at their temporary shapes was ultramicrotomed into thin pieces along the longitudinal deforming direction to conduct TEM observation. The EPDM/PP interface played three critical roles: transferring the stress from PP phase to EPDM particles in shaping process, keeping the deformation of EPDM particles in temporary shape and again transferring the stress from EPDM particles to PP phase in shape recovery process. When stretching, the thickness and width of parallel area of the middle part of dumbbell-shaped samples were continuously decreased. The orientation of PP phase forced the EPDM particle to be elongated and oriented along the stretching direction via the enhanced EPDM/PP interface. As shown in Figure 10, the EPDM particles can maintain their highly elongated shape in ultrathin section, strongly confirming a powerful compatibilized EPDM/PP interface. Furthermore, the EPDM particles were elongated to be “nano-fibers”, showing the trend of approaching each other. This is suggested that, once the strength of PP continuous phase was sharply decreased at trigger temperature, the elongated and oriented EPDM “nano-fibers” released strong elastic force to fulfill the shape recovery. However, it should be pointed out that the shape memory behavior was not ideal at lower EPDM (EPDM/PP

at 50/50, 40/60, ...) content due to that the reduced EPDM phase could not provide sufficient elastic restoring force for shape recovery. Of course, the shape fixity of PP-rich TPVs was remarkably improved. The repeated SM behaviors of TPVs were also evaluated by a stress-controlled cyclic thermal mechanical experiment which conducted at a T_d 120°C and a T_s 165°C (Supporting Information, Figure S5). The close SM cycles of the E6P4-2:1.3 indicated an excellent SM performance due to the reinforcement of ZnO nanoparticles on both EPDM and PP phases.

4. Conclusion

Considerable SM effect of EPDM/PP TPVs with sea-island structure was successfully achieved by the in-situ compatibilization of MAA and excess ZnO nanoparticles. The enhanced interface improved the efficiency of stress delivery between PP and EPDM phases and played an important role in deforming EPDM particles during shape fixing. The PP phase served as a "container" to fix the deformed EPDM particles via the strong interface. The crosslinked EPDM particles provide strong elastic restoring force. ZnO nanoparticles served as nucleating agents to increase the crystallinity of PP phase, which was positive to improve the SF of TPVs. Furthermore, the residual ZnO nanoparticles had an additional reinforcement on the EPDM particles, which provided stronger resilience to fulfill the SR. The largely increased EPDM/PP contact areas due to the reduction of EPDM particle size also contributed to the SM effect. The suitable T_d and T_s were found to be 120°C and 165°C, respectively, for this particular TPVs system. Enhanced tensile strength (~9.1 MPa) and Young's modulus (~380 MPa), improved SF (~99%) and SR (~98%) of TPVs are achieved by tailoring the MAA/ZnO ratio (2:1.3). We envision that our work would open up new avenues to design sea-island structured SM blends whose components have no inherent SM characteristics.

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